Final paper results Sept 2

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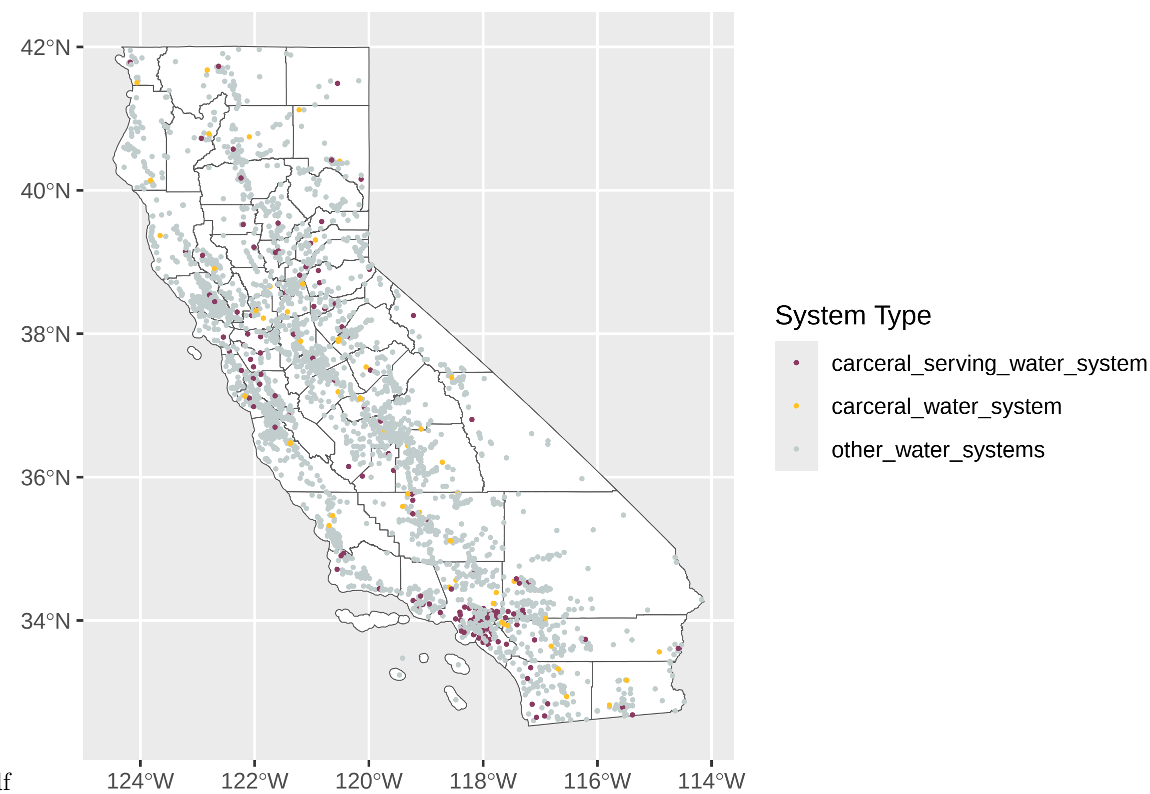
Data overview

Our data set includes 4260 unique public water systems representing all of the active CWS and NTNCWS water systems in the same. Of these water systems 2812 are community water systems, 1448 are Non-Transient Non-Community systems. The mean population served by these systems is 9583.4584507.

Describing carceral water systems

We classify these systems into three categories: Carceral water systems that primarily or exclusively serve carceral facilities; non-exclusive carceral water systems that serve one or more carceral facilities in addition to other customers and non-carceral systems that do not serve any carceral faciltiies. We find that 56 or 1.314554% are carceral water systems. 152 or 3.5680751% are non-exclusive carceral systems and 4052 or 95.1173709% are noncarceral systems.

As you can see above, many of the carceral serving systems rely on surface water. Which is interesting. Many of those systems are concentrated in the bay area, Sacramento and northern central valley areas and southern california whereas exclusive carceral systems are fairly evenly spread throughout the state.



map-1.pdf

Characteristic	carceral_serving_water_system N = 152[1]	carceral_water_system
pws_type_code		
CWS	147 (97%)	44 (79%)
NTNCWS	5 (3.3%)	12 (21%)
population_served_count	63,982 (16,057, 137,023)	533 (100, 3,729)
watersource		
GW	44 (29%)	45 (80%)
SW	108 (71%)	11 (20%)
purchased		
Not purchased	94 (62%)	51 (91%)
Purchased	58 (38%)	5 (8.9%)
HR_NAME		
Central Coast	9 (6.0%)	6 (11%)
Colorado River	4 (2.7%)	4 (7.4%)
North Coast	6 (4.0%)	5 (9.3%)
North Lahontan	4 (2.7%)	1 (1.9%)
Sacramento River	18 (12%)	8 (15%)
San Francisco Bay	15 (10%)	1 (1.9%)
San Joaquin River	11 (7.3%)	7 (13%)
South Coast	67 (45%)	9 (17%)
South Lahontan	5 (3.3%)	3 (5.6%)
Tulare Lake	11 (7.3%)	10 (19%)
Unknown	2	2

[1] n (%); Median (Q1, Q3)

Outcome (performance) descriptive results

Next we can see if these categories of water systems perform differently using different performance measures related to water quality, accessibility and TMF capacity.

GLMs for outcomes with controls

Lastly we can use GLMs to see if any of these relationships are stastisticallly significant while holding other driving factors like system size constant.

```
Data$carceral_sys <- relevel(Data$carceral_sys, ref = "other_water_systems")

MCL_violations <- glm(health_5yr_no2ndary_or_lsl_binary ~ carceral_sys + population_served_count + wate
summary(MCL_violations)
```

```
##
## Call:
## glm(formula = health_5yr_no2ndary_or_lsl_binary ~ carceral_sys +
##     population_served_count + watersource + purchased + pws_type_code +
##     HR_NAME, family = binomial, data = Data)
##
## Coefficients:
##                                              Estimate Std. Error z value
```

```
## (Intercept)                              -1.668e+00  1.292e-01 -12.917
## carceral_syscarceral_serving_water_system  1.098e-01  3.169e-01   0.347
## carceral_syscarceral_water_system         -3.303e-01  4.170e-01  -0.792
## population_served_count                   -4.688e-06  2.581e-06  -1.816
## watersourceSW                              3.908e-01  1.448e-01   2.700
## purchasedPurchased                        -1.217e+00  2.435e-01  -4.999
## pws_type_codeNTNCWS                       -1.303e-01  1.227e-01  -1.062
## HR_NAMEColorado River                      6.314e-02  2.832e-01   0.223
## HR_NAMENorth Coast                        -5.435e-01  2.177e-01  -2.497
## HR_NAMENorth Lahontan                     -5.799e-01  4.483e-01  -1.294
## HR_NAMESacramento River                   -7.596e-01  2.077e-01  -3.657
## HR_NAMESan Francisco Bay                  -8.018e-01  3.082e-01  -2.601
## HR_NAMESan Joaquin River                   4.766e-01  1.629e-01   2.926
## HR_NAMESouth Coast                        -1.309e-01  2.157e-01  -0.607
## HR_NAMESouth Lahontan                      1.223e-01  2.202e-01   0.556
## HR_NAMETulare Lake                         8.724e-01  1.623e-01   5.377
##                                           Pr(>|z|)
## (Intercept)                                < 2e-16 ***
## carceral_syscarceral_serving_water_system 0.728966
## carceral_syscarceral_water_system         0.428387
## population_served_count                   0.069315 .
## watersourceSW                             0.006944 **
## purchasedPurchased                        5.76e-07 ***
## pws_type_codeNTNCWS                       0.288260
## HR_NAMEColorado River                     0.823566
## HR_NAMENorth Coast                        0.012521 *
## HR_NAMENorth Lahontan                     0.195779
## HR_NAMESacramento River                   0.000255 ***
## HR_NAMESan Francisco Bay                  0.009289 **
## HR_NAMESan Joaquin River                  0.003437 **
## HR_NAMESouth Coast                        0.543887
## HR_NAMESouth Lahontan                     0.578512
## HR_NAMETulare Lake                        7.57e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 2974.4  on 3442  degrees of freedom
## Residual deviance: 2791.6  on 3427  degrees of freedom
##   (817 observations deleted due to missingness)
## AIC: 2823.6
##
## Number of Fisher Scoring iterations: 6
```

```r
M_violations <- glm(mon_only_5yr_binary ~ carceral_sys + population_served_count + watersource + purcha
summary(M_violations)
```

```
##
## Call:
## glm(formula = mon_only_5yr_binary ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, family = binomial,
##     data = Data)
##
## Coefficients:
```

```
##                                        Estimate Std. Error z value
## (Intercept)                           -6.106e-01  9.891e-02  -6.173
## carceral_syscarceral_serving_water_system -2.030e-01  2.319e-01  -0.875
## carceral_syscarceral_water_system      2.813e-01  2.840e-01   0.991
## population_served_count               -1.941e-06  1.386e-06  -1.401
## watersourceSW                         -3.500e-01  1.173e-01  -2.984
## purchasedPurchased                    -2.813e-01  1.532e-01  -1.837
## pws_type_codeNTNCWS                   -3.061e-01  9.913e-02  -3.088
## HR_NAMEColorado River                  1.030e+00  2.117e-01   4.866
## HR_NAMENorth Coast                     1.404e-01  1.473e-01   0.953
## HR_NAMENorth Lahontan                 -5.225e-01  3.188e-01  -1.639
## HR_NAMESacramento River                2.095e-02  1.368e-01   0.153
## HR_NAMESan Francisco Bay              -2.670e-01  1.930e-01  -1.384
## HR_NAMESan Joaquin River              -1.984e-01  1.362e-01  -1.456
## HR_NAMESouth Coast                     2.030e-01  1.524e-01   1.333
## HR_NAMESouth Lahontan                  4.299e-01  1.647e-01   2.611
## HR_NAMETulare Lake                    -3.130e-02  1.394e-01  -0.225
##                                       Pr(>|z|)
## (Intercept)                           6.69e-10 ***
## carceral_syscarceral_serving_water_system  0.38137
## carceral_syscarceral_water_system      0.32190
## population_served_count                0.16136
## watersourceSW                          0.00284 **
## purchasedPurchased                     0.06628 .
## pws_type_codeNTNCWS                    0.00201 **
## HR_NAMEColorado River                 1.14e-06 ***
## HR_NAMENorth Coast                     0.34051
## HR_NAMENorth Lahontan                  0.10128
## HR_NAMESacramento River                0.87833
## HR_NAMESan Francisco Bay               0.16647
## HR_NAMESan Joaquin River               0.14527
## HR_NAMESouth Coast                     0.18268
## HR_NAMESouth Lahontan                  0.00904 **
## HR_NAMETulare Lake                     0.82234
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 4327.5  on 3442  degrees of freedom
## Residual deviance: 4224.3  on 3427  degrees of freedom
##   (817 observations deleted due to missingness)
## AIC: 4256.3
##
## Number of Fisher Scoring iterations: 5
```

```r
MR_violations <- glm(mr_5yr_binary ~ carceral_sys + population_served_count + watersource + purchased +
summary(MR_violations)
```

```
##
## Call:
## glm(formula = mr_5yr_binary ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, family = binomial,
##     data = Data)
##
```

```
## Coefficients:
##                                             Estimate Std. Error z value
## (Intercept)                               -8.650e-02  9.347e-02  -0.925
## carceral_syscarceral_serving_water_system -1.722e-01  2.126e-01  -0.810
## carceral_syscarceral_water_system          2.297e-01  2.796e-01   0.822
## population_served_count                   -2.794e-06  1.357e-06  -2.059
## watersourceSW                             -4.405e-01  1.084e-01  -4.066
## purchasedPurchased                        -2.573e-01  1.403e-01  -1.834
## pws_type_codeNTNCWS                       -2.298e-01  9.099e-02  -2.525
## HR_NAMEColorado River                      1.083e+00  2.203e-01   4.916
## HR_NAMENorth Coast                         1.836e-01  1.398e-01   1.314
## HR_NAMENorth Lahontan                     -6.370e-01  2.901e-01  -2.196
## HR_NAMESacramento River                    3.320e-01  1.290e-01   2.573
## HR_NAMESan Francisco Bay                  -3.871e-02  1.722e-01  -0.225
## HR_NAMESan Joaquin River                  -1.391e-01  1.264e-01  -1.101
## HR_NAMESouth Coast                         1.345e-01  1.448e-01   0.929
## HR_NAMESouth Lahontan                      2.576e-01  1.602e-01   1.608
## HR_NAMETulare Lake                         1.085e-01  1.306e-01   0.831
##                                           Pr(>|z|)
## (Intercept)                                 0.3547
## carceral_syscarceral_serving_water_system   0.4181
## carceral_syscarceral_water_system           0.4113
## population_served_count                     0.0395 *
## watersourceSW                             4.79e-05 ***
## purchasedPurchased                          0.0666 .
## pws_type_codeNTNCWS                         0.0116 *
## HR_NAMEColorado River                     8.83e-07 ***
## HR_NAMENorth Coast                          0.1888
## HR_NAMENorth Lahontan                       0.0281 *
## HR_NAMESacramento River                     0.0101 *
## HR_NAMESan Francisco Bay                    0.8221
## HR_NAMESan Joaquin River                    0.2710
## HR_NAMESouth Coast                          0.3531
## HR_NAMESouth Lahontan                       0.1079
## HR_NAMETulare Lake                          0.4060
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 4747.0  on 3442  degrees of freedom
## Residual deviance: 4619.9  on 3427  degrees of freedom
##   (817 observations deleted due to missingness)
## AIC: 4651.9
##
## Number of Fisher Scoring iterations: 5
```

```r
NA_WQ <- lm(Water_Quality_Category_Score ~ carceral_sys + population_served_count + watersource + purcha
summary(NA_WQ)
```

```
##
## Call:
## lm(formula = Water_Quality_Category_Score ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, data = Data,
##     family = binomial)
```

```
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.3139 -0.6792 -0.3917  0.5941  5.1570
##
## Coefficients:
##                                             Estimate Std. Error t value
## (Intercept)                                8.427e-01  4.952e-02  17.017
## carceral_syscarceral_serving_water_system  4.374e-02  1.155e-01   0.379
## carceral_syscarceral_water_system          3.541e-02  1.522e-01   0.233
## population_served_count                    2.902e-06  1.546e-06   1.877
## watersourceSW                             -2.015e-01  5.662e-02  -3.559
## purchasedPurchased                        -2.474e-01  7.266e-02  -3.405
## pws_type_codeNTNCWS                        1.656e-02  7.397e-02   0.224
## HR_NAMEColorado River                      1.354e-01  1.092e-01   1.240
## HR_NAMENorth Coast                        -4.410e-01  7.641e-02  -5.772
## HR_NAMENorth Lahontan                     -5.130e-01  1.445e-01  -3.549
## HR_NAMESacramento River                   -2.997e-01  6.762e-02  -4.432
## HR_NAMESan Francisco Bay                  -3.064e-01  9.616e-02  -3.186
## HR_NAMESan Joaquin River                   2.108e-02  6.822e-02   0.309
## HR_NAMESouth Coast                         1.232e-01  7.537e-02   1.634
## HR_NAMESouth Lahontan                      6.209e-02  8.549e-02   0.726
## HR_NAMETulare Lake                         3.522e-01  7.188e-02   4.900
##                                           Pr(>|t|)
## (Intercept)                                < 2e-16 ***
## carceral_syscarceral_serving_water_system 0.704989
## carceral_syscarceral_water_system         0.816102
## population_served_count                   0.060647 .
## watersourceSW                             0.000378 ***
## purchasedPurchased                        0.000671 ***
## pws_type_codeNTNCWS                       0.822814
## HR_NAMEColorado River                     0.215000
## HR_NAMENorth Coast                        8.69e-09 ***
## HR_NAMENorth Lahontan                     0.000393 ***
## HR_NAMESacramento River                   9.67e-06 ***
## HR_NAMESan Francisco Bay                  0.001458 **
## HR_NAMESan Joaquin River                  0.757285
## HR_NAMESouth Coast                        0.102340
## HR_NAMESouth Lahontan                     0.467700
## HR_NAMETulare Lake                        1.01e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9985 on 2853 degrees of freedom
##   (1391 observations deleted due to missingness)
## Multiple R-squared:  0.07613,    Adjusted R-squared:  0.07127
## F-statistic: 15.67 on 15 and 2853 DF,  p-value: < 2.2e-16
```

```r
NA_AC <- lm(Accessibility_Category_Score ~ carceral_sys + population_served_count + watersource + purcha
summary(NA_AC)
```

```
##
## Call:
## lm(formula = Accessibility_Category_Score ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, data = Data,
```

```
##     family = binomial)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.5043 -0.5529 -0.1180  0.4474  3.4728
##
## Coefficients:
##                                          Estimate Std. Error t value
## (Intercept)                             9.359e-01  3.549e-02  26.375
## carceral_syscarceral_serving_water_system -4.781e-02  8.279e-02  -0.577
## carceral_syscarceral_water_system        -9.733e-02  1.091e-01  -0.892
## population_served_count                 -1.216e-05  1.108e-06 -10.978
## watersourceSW                            3.469e-02  4.057e-02   0.855
## purchasedPurchased                      -2.245e-01  5.207e-02  -4.312
## pws_type_codeNTNCWS                      5.716e-01  5.300e-02  10.784
## HR_NAMEColorado River                   -8.721e-02  7.825e-02  -1.114
## HR_NAMENorth Coast                      -6.704e-02  5.476e-02  -1.224
## HR_NAMENorth Lahontan                   -4.191e-01  1.036e-01  -4.047
## HR_NAMESacramento River                  3.350e-02  4.846e-02   0.691
## HR_NAMESan Francisco Bay                -1.185e-01  6.890e-02  -1.719
## HR_NAMESan Joaquin River                 1.617e-01  4.888e-02   3.309
## HR_NAMESouth Coast                      -2.167e-01  5.401e-02  -4.012
## HR_NAMESouth Lahontan                   -1.633e-01  6.126e-02  -2.665
## HR_NAMETulare Lake                       4.722e-01  5.151e-02   9.168
##                                         Pr(>|t|)
## (Intercept)                              < 2e-16 ***
## carceral_syscarceral_serving_water_system 0.563663
## carceral_syscarceral_water_system        0.372352
## population_served_count                  < 2e-16 ***
## watersourceSW                           0.392558
## purchasedPurchased                      1.67e-05 ***
## pws_type_codeNTNCWS                      < 2e-16 ***
## HR_NAMEColorado River                   0.265176
## HR_NAMENorth Coast                      0.220919
## HR_NAMENorth Lahontan                   5.33e-05 ***
## HR_NAMESacramento River                 0.489372
## HR_NAMESan Francisco Bay                0.085689 .
## HR_NAMESan Joaquin River                0.000949 ***
## HR_NAMESouth Coast                      6.17e-05 ***
## HR_NAMESouth Lahontan                   0.007732 **
## HR_NAMETulare Lake                       < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7155 on 2853 degrees of freedom
##   (1391 observations deleted due to missingness)
## Multiple R-squared:  0.2212, Adjusted R-squared:  0.2171
## F-statistic: 54.03 on 15 and 2853 DF,  p-value: < 2.2e-16
```

```
NA_TMF <- lm(TMF_Capacity_Category_Score ~ carceral_sys + population_served_count + watersource + purcha
summary(NA_TMF)
```

```
##
## Call:
## lm(formula = TMF_Capacity_Category_Score ~ carceral_sys + population_served_count +
```

```
##     watersource + purchased + pws_type_code + HR_NAME, data = Data,
##     family = binomial)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.79161 -0.37214 -0.05872  0.26137  2.16083
##
## Coefficients:
##                                             Estimate Std. Error t value
## (Intercept)                                3.724e-01  2.074e-02  17.957
## carceral_syscarceral_serving_water_system -1.443e-02  4.838e-02  -0.298
## carceral_syscarceral_water_system          5.048e-01  6.375e-02   7.918
## population_served_count                   -3.331e-06  6.474e-07  -5.145
## watersourceSW                              2.248e-02  2.371e-02   0.948
## purchasedPurchased                        -1.037e-01  3.043e-02  -3.407
## pws_type_codeNTNCWS                       -4.044e-01  3.097e-02 -13.057
## HR_NAMEColorado River                      2.711e-01  4.573e-02   5.928
## HR_NAMENorth Coast                         3.473e-02  3.200e-02   1.085
## HR_NAMENorth Lahontan                      1.884e-02  6.052e-02   0.311
## HR_NAMESacramento River                    8.638e-02  2.832e-02   3.050
## HR_NAMESan Francisco Bay                   2.553e-02  4.027e-02   0.634
## HR_NAMESan Joaquin River                   9.493e-02  2.857e-02   3.323
## HR_NAMESouth Coast                         1.401e-01  3.156e-02   4.439
## HR_NAMESouth Lahontan                      1.349e-01  3.580e-02   3.767
## HR_NAMETulare Lake                         1.242e-01  3.010e-02   4.126
##                                           Pr(>|t|)
## (Intercept)                                < 2e-16 ***
## carceral_syscarceral_serving_water_system 0.765596
## carceral_syscarceral_water_system         3.42e-15 ***
## population_served_count                   2.86e-07 ***
## watersourceSW                             0.343226
## purchasedPurchased                        0.000665 ***
## pws_type_codeNTNCWS                        < 2e-16 ***
## HR_NAMEColorado River                     3.43e-09 ***
## HR_NAMENorth Coast                        0.277878
## HR_NAMENorth Lahontan                     0.755593
## HR_NAMESacramento River                   0.002306 **
## HR_NAMESan Francisco Bay                  0.526038
## HR_NAMESan Joaquin River                  0.000902 ***
## HR_NAMESouth Coast                        9.39e-06 ***
## HR_NAMESouth Lahontan                     0.000168 ***
## HR_NAMETulare Lake                        3.80e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4181 on 2853 degrees of freedom
##   (1391 observations deleted due to missingness)
## Multiple R-squared:  0.1122, Adjusted R-squared:  0.1075
## F-statistic: 24.04 on 15 and 2853 DF,  p-value: < 2.2e-16
```

```r
Data$Failingoratrisk <- ifelse(Data$SAFER_Status == "Failing" | Data$SAFER_Status == "At-Risk", 1, 0)
Data$failing <- ifelse(Data$SAFER_Status == "Failing", 1, 0)

Fail_or_atrisk <- lm(Failingoratrisk ~ carceral_sys + population_served_count + watersource + purchased
```

```r
summary(Fail_or_atrisk)
```

```
##
## Call:
## lm(formula = Failingoratrisk ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, data = Data,
##     family = binomial)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.6513 -0.3170 -0.2273  0.5389  1.0403
##
## Coefficients:
##                                             Estimate Std. Error t value
## (Intercept)                                3.172e-01  2.218e-02  14.303
## carceral_syscarceral_serving_water_system  1.994e-02  5.174e-02   0.385
## carceral_syscarceral_water_system          6.079e-02  6.818e-02   0.892
## population_served_count                   -3.060e-06  6.924e-07  -4.420
## watersourceSW                             -7.738e-03  2.536e-02  -0.305
## purchasedPurchased                        -1.203e-01  3.254e-02  -3.695
## pws_type_codeNTNCWS                       -1.370e-02  3.313e-02  -0.414
## HR_NAMEColorado River                      1.359e-01  4.891e-02   2.778
## HR_NAMENorth Coast                        -7.604e-02  3.422e-02  -2.222
## HR_NAMENorth Lahontan                     -2.156e-01  6.473e-02  -3.330
## HR_NAMESacramento River                   -5.740e-02  3.029e-02  -1.895
## HR_NAMESan Francisco Bay                  -9.728e-02  4.307e-02  -2.259
## HR_NAMESan Joaquin River                   1.209e-01  3.055e-02   3.959
## HR_NAMESouth Coast                         3.494e-02  3.375e-02   1.035
## HR_NAMESouth Lahontan                      5.365e-02  3.829e-02   1.401
## HR_NAMETulare Lake                         2.734e-01  3.219e-02   8.494
##                                           Pr(>|t|)
## (Intercept)                                < 2e-16 ***
## carceral_syscarceral_serving_water_system 0.699966
## carceral_syscarceral_water_system         0.372710
## population_served_count                   1.02e-05 ***
## watersourceSW                             0.760274
## purchasedPurchased                        0.000224 ***
## pws_type_codeNTNCWS                       0.679147
## HR_NAMEColorado River                     0.005502 **
## HR_NAMENorth Coast                        0.026361 *
## HR_NAMENorth Lahontan                     0.000878 ***
## HR_NAMESacramento River                   0.058175 .
## HR_NAMESan Francisco Bay                  0.023962 *
## HR_NAMESan Joaquin River                  7.72e-05 ***
## HR_NAMESouth Coast                        0.300665
## HR_NAMESouth Lahontan                     0.161245
## HR_NAMETulare Lake                         < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4472 on 2853 degrees of freedom
##   (1391 observations deleted due to missingness)
## Multiple R-squared:  0.0912, Adjusted R-squared:  0.08642
## F-statistic: 19.09 on 15 and 2853 DF,  p-value: < 2.2e-16
```

```r
Fail <- lm(failing ~ carceral_sys + population_served_count + watersource + purchased + pws_type_code +
summary(Fail)
```

```
##
## Call:
## lm(formula = failing ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, data = Data,
##     family = binomial)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.29542 -0.16479 -0.09601 -0.05269  0.99672
##
## Coefficients:
##                                             Estimate Std. Error t value
## (Intercept)                                1.156e-01  1.625e-02   7.115
## carceral_syscarceral_serving_water_system -2.158e-02  3.791e-02  -0.569
## carceral_syscarceral_water_system         -6.891e-02  4.995e-02  -1.380
## population_served_count                   -9.873e-07  5.073e-07  -1.946
## watersourceSW                             -1.885e-02  1.858e-02  -1.015
## purchasedPurchased                        -6.398e-02  2.384e-02  -2.684
## pws_type_codeNTNCWS                       -1.789e-02  2.427e-02  -0.737
## HR_NAMEColorado River                      9.921e-02  3.583e-02   2.769
## HR_NAMENorth Coast                        -1.932e-02  2.507e-02  -0.770
## HR_NAMENorth Lahontan                     -6.837e-02  4.742e-02  -1.442
## HR_NAMESacramento River                   -4.087e-02  2.219e-02  -1.842
## HR_NAMESan Francisco Bay                  -2.941e-02  3.155e-02  -0.932
## HR_NAMESan Joaquin River                   6.735e-02  2.238e-02   3.009
## HR_NAMESouth Coast                         3.567e-02  2.473e-02   1.442
## HR_NAMESouth Lahontan                      8.843e-02  2.805e-02   3.153
## HR_NAMETulare Lake                         1.798e-01  2.358e-02   7.625
##                                           Pr(>|t|)
## (Intercept)                               1.41e-12 ***
## carceral_syscarceral_serving_water_system  0.56929
## carceral_syscarceral_water_system          0.16785
## population_served_count                    0.05172 .
## watersourceSW                              0.31027
## purchasedPurchased                         0.00732 **
## pws_type_codeNTNCWS                        0.46118
## HR_NAMEColorado River                      0.00566 **
## HR_NAMENorth Coast                         0.44109
## HR_NAMENorth Lahontan                      0.14950
## HR_NAMESacramento River                    0.06557 .
## HR_NAMESan Francisco Bay                   0.35139
## HR_NAMESan Joaquin River                   0.00264 **
## HR_NAMESouth Coast                         0.14933
## HR_NAMESouth Lahontan                      0.00163 **
## HR_NAMETulare Lake                        3.30e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3276 on 2853 degrees of freedom
##   (1391 observations deleted due to missingness)
## Multiple R-squared:  0.05846,    Adjusted R-squared:  0.05351
```

```
## F-statistic: 11.81 on 15 and 2853 DF,  p-value: < 2.2e-16
```

```r
CCR_anymissing <- glm(Missing_ccr_any ~ carceral_sys + population_served_count + watersource + purchase
summary(CCR_anymissing)
```

```
##
## Call:
## glm(formula = Missing_ccr_any ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, family = binomial,
##     data = Data)
##
## Coefficients:
##                                             Estimate Std. Error z value
## (Intercept)                               -9.361e-01  1.057e-01  -8.855
## carceral_syscarceral_serving_water_system -2.339e-01  2.454e-01  -0.953
## carceral_syscarceral_water_system         -1.501e-01  3.210e-01  -0.467
## population_served_count                   -6.165e-06  1.869e-06  -3.298
## watersourceSW                             -3.808e-01  1.217e-01  -3.128
## purchasedPurchased                        -2.827e-01  1.549e-01  -1.825
## pws_type_codeNTNCWS                       -1.993e-01  1.056e-01  -1.888
## HR_NAMEColorado River                      1.048e+00  2.147e-01   4.881
## HR_NAMENorth Coast                         2.113e+00  1.587e-01  13.315
## HR_NAMENorth Lahontan                      3.007e+00  3.934e-01   7.645
## HR_NAMESacramento River                    2.290e+00  1.495e-01  15.312
## HR_NAMESan Francisco Bay                   1.838e+00  1.825e-01  10.072
## HR_NAMESan Joaquin River                   6.335e-01  1.360e-01   4.657
## HR_NAMESouth Coast                         6.832e-01  1.570e-01   4.352
## HR_NAMESouth Lahontan                     -3.668e-01  1.961e-01  -1.870
## HR_NAMETulare Lake                        -1.152e+00  1.869e-01  -6.167
##                                           Pr(>|z|)
## (Intercept)                                < 2e-16 ***
## carceral_syscarceral_serving_water_system 0.340613
## carceral_syscarceral_water_system         0.640204
## population_served_count                   0.000973 ***
## watersourceSW                             0.001758 **
## purchasedPurchased                        0.068038 .
## pws_type_codeNTNCWS                       0.059047 .
## HR_NAMEColorado River                     1.06e-06 ***
## HR_NAMENorth Coast                         < 2e-16 ***
## HR_NAMENorth Lahontan                     2.09e-14 ***
## HR_NAMESacramento River                    < 2e-16 ***
## HR_NAMESan Francisco Bay                   < 2e-16 ***
## HR_NAMESan Joaquin River                  3.21e-06 ***
## HR_NAMESouth Coast                        1.35e-05 ***
## HR_NAMESouth Lahontan                     0.061453 .
## HR_NAMETulare Lake                        6.98e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 4690.8  on 3442  degrees of freedom
## Residual deviance: 3769.6  on 3427  degrees of freedom
##   (817 observations deleted due to missingness)
## AIC: 3801.6
```

```
##
## Number of Fisher Scoring iterations: 6
```

```
CCR_missingtwoplus <- glm(Missing_ccr_twoplus ~ carceral_sys + population_served_count + watersource + p
summary(CCR_missingtwoplus)
```

```
##
## Call:
## glm(formula = Missing_ccr_twoplus ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, family = binomial,
##     data = Data)
##
## Coefficients:
##                                              Estimate Std. Error z value
## (Intercept)                                -1.591e+00  1.298e-01 -12.252
## carceral_syscarceral_serving_water_system -7.261e-01  3.895e-01  -1.864
## carceral_syscarceral_water_system         -1.374e+00  4.704e-01  -2.921
## population_served_count                   -3.416e-05  6.134e-06  -5.570
## watersourceSW                             -2.146e-01  1.306e-01  -1.643
## purchasedPurchased                        -3.922e-01  1.929e-01  -2.033
## pws_type_codeNTNCWS                       -2.426e-01  1.142e-01  -2.125
## HR_NAMEColorado River                      9.760e-01  2.464e-01   3.961
## HR_NAMENorth Coast                         1.833e+00  1.669e-01  10.979
## HR_NAMENorth Lahontan                      3.050e+00  3.329e-01   9.161
## HR_NAMESacramento River                    2.344e+00  1.610e-01  14.560
## HR_NAMESan Francisco Bay                   1.849e+00  2.007e-01   9.210
## HR_NAMESan Joaquin River                   8.693e-01  1.598e-01   5.439
## HR_NAMESouth Coast                         1.927e-01  2.108e-01   0.914
## HR_NAMESouth Lahontan                     -1.238e+00  3.349e-01  -3.696
## HR_NAMETulare Lake                        -1.692e+00  3.003e-01  -5.633
##                                            Pr(>|z|)
## (Intercept)                                 < 2e-16 ***
## carceral_syscarceral_serving_water_system 0.062273 .
## carceral_syscarceral_water_system         0.003489 **
## population_served_count                   2.55e-08 ***
## watersourceSW                             0.100405
## purchasedPurchased                        0.042059 *
## pws_type_codeNTNCWS                       0.033588 *
## HR_NAMEColorado River                     7.45e-05 ***
## HR_NAMENorth Coast                         < 2e-16 ***
## HR_NAMENorth Lahontan                      < 2e-16 ***
## HR_NAMESacramento River                    < 2e-16 ***
## HR_NAMESan Francisco Bay                   < 2e-16 ***
## HR_NAMESan Joaquin River                  5.37e-08 ***
## HR_NAMESouth Coast                        0.360510
## HR_NAMESouth Lahontan                     0.000219 ***
## HR_NAMETulare Lake                        1.77e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 4073.5  on 3442  degrees of freedom
## Residual deviance: 3096.7  on 3427  degrees of freedom
##   (817 observations deleted due to missingness)
```

```
## AIC: 3128.7
##
## Number of Fisher Scoring iterations: 8
```

```
Modeleddroughtimpact <- glm(SC5e_Drought_Impact ~ carceral_sys + population_served_count + watersource +
summary(Modeleddroughtimpact)
```

```
##
## Call:
## glm(formula = SC5e_Drought_Impact ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, family = binomial,
##     data = Data)
##
## Coefficients:
##                                            Estimate Std. Error z value
## (Intercept)                              -3.711e+00  3.276e-01 -11.328
## carceral_syscarceral_serving_water_system  6.196e-01  6.174e-01   1.003
## carceral_syscarceral_water_system        -1.632e+01  1.381e+03  -0.012
## population_served_count                   3.808e-06  3.074e-05   0.124
## watersourceSW                             1.482e+00  2.077e-01   7.138
## purchasedPurchased                       -1.173e+00  4.436e-01  -2.644
## pws_type_codeNTNCWS                      -1.663e+01  4.160e+02  -0.040
## HR_NAMEColorado River                     8.628e-02  6.787e-01   0.127
## HR_NAMENorth Coast                        1.050e+00  3.957e-01   2.655
## HR_NAMENorth Lahontan                    -4.088e-01  1.071e+00  -0.382
## HR_NAMESacramento River                   1.035e+00  3.703e-01   2.796
## HR_NAMESan Francisco Bay                 -1.246e-01  6.143e-01  -0.203
## HR_NAMESan Joaquin River                  1.558e-01  4.268e-01   0.365
## HR_NAMESouth Coast                       -4.914e-01  6.119e-01  -0.803
## HR_NAMESouth Lahontan                     4.703e-01  5.062e-01   0.929
## HR_NAMETulare Lake                        1.179e+00  3.904e-01   3.020
##                                          Pr(>|z|)
## (Intercept)                               < 2e-16 ***
## carceral_syscarceral_serving_water_system  0.31562
## carceral_syscarceral_water_system         0.99057
## population_served_count                   0.90142
## watersourceSW                            9.48e-13 ***
## purchasedPurchased                        0.00819 **
## pws_type_codeNTNCWS                       0.96810
## HR_NAMEColorado River                     0.89885
## HR_NAMENorth Coast                        0.00793 **
## HR_NAMENorth Lahontan                     0.70278
## HR_NAMESacramento River                   0.00518 **
## HR_NAMESan Francisco Bay                  0.83931
## HR_NAMESan Joaquin River                  0.71498
## HR_NAMESouth Coast                        0.42197
## HR_NAMESouth Lahontan                     0.35284
## HR_NAMETulare Lake                        0.00252 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1059.99  on 2884  degrees of freedom
## Residual deviance:  895.03  on 2869  degrees of freedom
```

```
##   (1375 observations deleted due to missingness)
## AIC: 927.03
##
## Number of Fisher Scoring iterations: 18
```

```
Distribution_disrupt <- glm(Distributiondis_any ~ carceral_sys + population_served_count + watersource +
summary(Distribution_disrupt)
```

```
##
## Call:
## glm(formula = Distributiondis_any ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, family = binomial,
##     data = Data)
##
## Coefficients:
##                                          Estimate Std. Error z value
## (Intercept)                            -1.960e+00  1.582e-01 -12.391
## carceral_syscarceral_serving_water_system  4.522e-01  4.902e-01   0.922
## carceral_syscarceral_water_system        -1.856e+00  1.016e+00  -1.827
## population_served_count                 1.789e-05  9.631e-06   1.857
## watersourceSW                           5.269e-01  1.658e-01   3.179
## purchasedPurchased                      5.217e-02  2.315e-01   0.225
## pws_type_codeNTNCWS                    -1.763e+00  2.765e-01  -6.375
## HR_NAMEColorado River                  -1.051e-01  3.599e-01  -0.292
## HR_NAMENorth Coast                      8.014e-02  2.364e-01   0.339
## HR_NAMENorth Lahontan                  -4.134e-03  4.670e-01  -0.009
## HR_NAMESacramento River                 9.002e-02  2.115e-01   0.426
## HR_NAMESan Francisco Bay                1.844e-01  2.874e-01   0.642
## HR_NAMESan Joaquin River                3.116e-02  2.143e-01   0.145
## HR_NAMESouth Coast                     -1.431e-01  2.570e-01  -0.557
## HR_NAMESouth Lahontan                  -6.206e-01  3.207e-01  -1.935
## HR_NAMETulare Lake                     -6.229e-01  2.718e-01  -2.292
##                                        Pr(>|z|)
## (Intercept)                             < 2e-16 ***
## carceral_syscarceral_serving_water_system  0.35628
## carceral_syscarceral_water_system         0.06764 .
## population_served_count                  0.06330 .
## watersourceSW                            0.00148 **
## purchasedPurchased                       0.82168
## pws_type_codeNTNCWS                     1.83e-10 ***
## HR_NAMEColorado River                    0.77033
## HR_NAMENorth Coast                       0.73459
## HR_NAMENorth Lahontan                    0.99294
## HR_NAMESacramento River                  0.67041
## HR_NAMESan Francisco Bay                 0.52118
## HR_NAMESan Joaquin River                 0.88440
## HR_NAMESouth Coast                       0.57760
## HR_NAMESouth Lahontan                    0.05302 .
## HR_NAMETulare Lake                       0.02193 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1947.3  on 2847  degrees of freedom
```

```
## Residual deviance: 1824.2  on 2832  degrees of freedom
##   (1412 observations deleted due to missingness)
## AIC: 1856.2
##
## Number of Fisher Scoring iterations: 6
```

```r
Sig_dif <- glm(Significant_Deficiencies ~ carceral_sys + population_served_count + watersource + purcha
summary(Sig_dif)
```

```
##
## Call:
## glm(formula = Significant_Deficiencies ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, family = binomial,
##     data = Data)
##
## Coefficients:
##                                            Estimate Std. Error z value
## (Intercept)                              -4.106e+00  4.145e-01  -9.906
## carceral_syscarceral_serving_water_system -1.781e+01  2.768e+03  -0.006
## carceral_syscarceral_water_system          1.162e+00  7.753e-01   1.499
## population_served_count                    2.247e-05  1.160e-05   1.937
## watersourceSW                             -1.675e+00  8.904e-01  -1.881
## purchasedPurchased                         2.271e-01  8.438e-01   0.269
## pws_type_codeNTNCWS                       -1.625e+01  1.819e+03  -0.009
## HR_NAMEColorado River                      7.109e-01  7.211e-01   0.986
## HR_NAMENorth Coast                        -1.251e+00  1.085e+00  -1.153
## HR_NAMENorth Lahontan                      1.068e+00  8.363e-01   1.277
## HR_NAMESacramento River                   -4.447e-01  6.507e-01  -0.683
## HR_NAMESan Francisco Bay                  -1.693e+01  2.255e+03  -0.008
## HR_NAMESan Joaquin River                  -1.718e+01  1.347e+03  -0.013
## HR_NAMESouth Coast                         8.736e-01  5.381e-01   1.624
## HR_NAMESouth Lahontan                      2.804e-01  6.538e-01   0.429
## HR_NAMETulare Lake                        -1.727e+01  1.481e+03  -0.012
##                                           Pr(>|z|)
## (Intercept)                                 <2e-16 ***
## carceral_syscarceral_serving_water_system   0.9949
## carceral_syscarceral_water_system           0.1339
## population_served_count                     0.0528 .
## watersourceSW                               0.0599 .
## purchasedPurchased                          0.7878
## pws_type_codeNTNCWS                         0.9929
## HR_NAMEColorado River                       0.3242
## HR_NAMENorth Coast                          0.2488
## HR_NAMENorth Lahontan                       0.2015
## HR_NAMESacramento River                     0.4943
## HR_NAMESan Francisco Bay                    0.9940
## HR_NAMESan Joaquin River                    0.9898
## HR_NAMESouth Coast                          0.1045
## HR_NAMESouth Lahontan                       0.6680
## HR_NAMETulare Lake                          0.9907
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
```

```
##     Null deviance: 351.38  on 2868  degrees of freedom
## Residual deviance: 296.24  on 2853  degrees of freedom
##   (1391 observations deleted due to missingness)
## AIC: 328.24
##
## Number of Fisher Scoring iterations: 20
```

```r
Bottledwaterreliance <- glm(Bottled_Water_or_Hauled_Water_Reliance ~ carceral_sys + population_served_c
summary(Bottledwaterreliance)
```

```
##
## Call:
## glm(formula = Bottled_Water_or_Hauled_Water_Reliance ~ carceral_sys +
##     population_served_count + watersource + purchased + pws_type_code +
##     HR_NAME, family = binomial, data = Data)
##
## Coefficients:
##                                              Estimate Std. Error z value
## (Intercept)                                -2.846e+00  2.175e-01 -13.086
## carceral_syscarceral_serving_water_system -3.453e-01  1.046e+00  -0.330
## carceral_syscarceral_water_system         -8.595e-01  1.018e+00  -0.844
## population_served_count                   -7.492e-05  2.562e-05  -2.924
## watersourceSW                               1.160e-01  2.688e-01   0.431
## purchasedPurchased                          2.723e-01  3.424e-01   0.795
## pws_type_codeNTNCWS                         1.137e+00  2.297e-01   4.948
## HR_NAMEColorado River                       9.713e-02  4.780e-01   0.203
## HR_NAMENorth Coast                         -4.600e-01  3.596e-01  -1.279
## HR_NAMENorth Lahontan                      -1.230e+00  1.036e+00  -1.188
## HR_NAMESacramento River                    -1.956e-01  3.144e-01  -0.622
## HR_NAMESan Francisco Bay                   -1.113e-01  4.505e-01  -0.247
## HR_NAMESan Joaquin River                   -1.691e-01  3.038e-01  -0.557
## HR_NAMESouth Coast                          1.375e-01  3.472e-01   0.396
## HR_NAMESouth Lahontan                       1.060e-01  3.682e-01   0.288
## HR_NAMETulare Lake                          4.050e-01  2.855e-01   1.419
##                                            Pr(>|z|)
## (Intercept)                                 < 2e-16 ***
## carceral_syscarceral_serving_water_system  0.74143
## carceral_syscarceral_water_system          0.39864
## population_served_count                    0.00345 **
## watersourceSW                              0.66619
## purchasedPurchased                         0.42646
## pws_type_codeNTNCWS                        7.48e-07 ***
## HR_NAMEColorado River                      0.83897
## HR_NAMENorth Coast                         0.20081
## HR_NAMENorth Lahontan                      0.23491
## HR_NAMESacramento River                    0.53385
## HR_NAMESan Francisco Bay                   0.80490
## HR_NAMESan Joaquin River                   0.57772
## HR_NAMESouth Coast                         0.69200
## HR_NAMESouth Lahontan                      0.77349
## HR_NAMETulare Lake                         0.15603
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
```

```
##
##     Null deviance: 1200.4  on 2868  degrees of freedom
## Residual deviance: 1139.1  on 2853  degrees of freedom
##   (1391 observations deleted due to missingness)
## AIC: 1171.1
##
## Number of Fisher Scoring iterations: 8
```

```r
#Last thing, try EPA health and monitoring binary
EPA <- read_csv("Data/usepa_df_for_analysis.csv")
EPA <- EPA[,c(2,11,12)]
Data <- left_join(Data, EPA)

MCL_violations_epa <- glm(binary_viol_over5yrs_health ~ carceral_sys + population_served_count + waters
summary(MCL_violations_epa)
```

```
##
## Call:
## glm(formula = binary_viol_over5yrs_health ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, family = binomial,
##     data = Data)
##
## Coefficients:
##                                              Estimate Std. Error z value
## (Intercept)                                -1.792e+00  1.354e-01 -13.232
## carceral_syscarceral_serving_water_system   1.809e-01  3.472e-01   0.521
## carceral_syscarceral_water_system          -5.888e-01  5.278e-01  -1.115
## population_served_count                    -5.802e-06  2.993e-06  -1.939
## watersourceSW                               6.267e-01  1.556e-01   4.029
## purchasedPurchased                         -1.514e+00  2.776e-01  -5.454
## pws_type_codeNTNCWS                        -1.622e-01  1.419e-01  -1.143
## HR_NAMEColorado River                       7.254e-02  2.921e-01   0.248
## HR_NAMENorth Coast                         -1.053e+00  2.628e-01  -4.006
## HR_NAMENorth Lahontan                      -1.657e+00  7.317e-01  -2.264
## HR_NAMESacramento River                    -9.254e-01  2.256e-01  -4.102
## HR_NAMESan Francisco Bay                   -5.040e-01  2.913e-01  -1.730
## HR_NAMESan Joaquin River                   -4.504e-02  1.823e-01  -0.247
## HR_NAMESouth Coast                         -1.957e-01  2.307e-01  -0.848
## HR_NAMESouth Lahontan                      -3.367e-02  2.393e-01  -0.141
## HR_NAMETulare Lake                          4.291e-01  1.765e-01   2.432
##                                            Pr(>|z|)
## (Intercept)                                 < 2e-16 ***
## carceral_syscarceral_serving_water_system    0.6024
## carceral_syscarceral_water_system            0.2647
## population_served_count                      0.0525 .
## watersourceSW                              5.61e-05 ***
## purchasedPurchased                         4.93e-08 ***
## pws_type_codeNTNCWS                          0.2529
## HR_NAMEColorado River                        0.8038
## HR_NAMENorth Coast                         6.19e-05 ***
## HR_NAMENorth Lahontan                        0.0236 *
## HR_NAMESacramento River                    4.09e-05 ***
## HR_NAMESan Francisco Bay                     0.0836 .
## HR_NAMESan Joaquin River                     0.8048
## HR_NAMESouth Coast                           0.3962
```

```
## HR_NAMESouth Lahontan                         0.8881
## HR_NAMETulare Lake                            0.0150 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 2437.0  on 3442  degrees of freedom
## Residual deviance: 2315.9  on 3427  degrees of freedom
##   (817 observations deleted due to missingness)
## AIC: 2347.9
##
## Number of Fisher Scoring iterations: 6
```

```
MR_violations_epa <- glm(binary_viol_over5yrs_mon ~ carceral_sys + population_served_count + watersourc
summary(MR_violations_epa)
```

```
##
## Call:
## glm(formula = binary_viol_over5yrs_mon ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, family = binomial,
##     data = Data)
##
## Coefficients:
##                                           Estimate Std. Error z value
## (Intercept)                              -1.002e+00  1.061e-01  -9.443
## carceral_syscarceral_serving_water_system -6.732e-02  2.541e-01  -0.265
## carceral_syscarceral_water_system         2.598e-01  2.997e-01   0.867
## population_served_count                  -4.130e-06  1.918e-06  -2.153
## watersourceSW                            -1.337e-01  1.253e-01  -1.067
## purchasedPurchased                       -3.397e-01  1.662e-01  -2.044
## pws_type_codeNTNCWS                      -9.648e-02  1.058e-01  -0.912
## HR_NAMEColorado River                     8.541e-01  2.145e-01   3.981
## HR_NAMENorth Coast                        2.857e-01  1.539e-01   1.857
## HR_NAMENorth Lahontan                    -2.909e-01  3.284e-01  -0.886
## HR_NAMESacramento River                  -1.400e-01  1.495e-01  -0.937
## HR_NAMESan Francisco Bay                 -1.505e-01  2.034e-01  -0.740
## HR_NAMESan Joaquin River                 -4.697e-01  1.529e-01  -3.073
## HR_NAMESouth Coast                        2.755e-01  1.623e-01   1.697
## HR_NAMESouth Lahontan                     2.769e-01  1.760e-01   1.574
## HR_NAMETulare Lake                       -1.644e-01  1.518e-01  -1.083
##                                          Pr(>|z|)
## (Intercept)                               < 2e-16 ***
## carceral_syscarceral_serving_water_system  0.79106
## carceral_syscarceral_water_system          0.38612
## population_served_count                    0.03129 *
## watersourceSW                              0.28601
## purchasedPurchased                         0.04095 *
## pws_type_codeNTNCWS                        0.36163
## HR_NAMEColorado River                    6.87e-05 ***
## HR_NAMENorth Coast                         0.06334 .
## HR_NAMENorth Lahontan                      0.37570
## HR_NAMESacramento River                    0.34900
## HR_NAMESan Francisco Bay                   0.45941
## HR_NAMESan Joaquin River                   0.00212 **
```

```
## HR_NAMESouth Coast                          0.08969 .
## HR_NAMESouth Lahontan                       0.11552
## HR_NAMETulare Lake                          0.27868
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 3839.5  on 3442  degrees of freedom
## Residual deviance: 3757.9  on 3427  degrees of freedom
##   (817 observations deleted due to missingness)
## AIC: 3789.9
##
## Number of Fisher Scoring iterations: 6
```